Integration and Kinematics

Section 1: Indefinite Integration Techniques

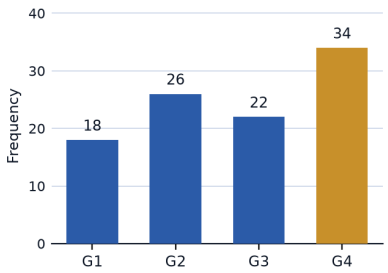
1 Find the indefinite integral $\int (3x-1)^4\,dx$.

a. $\frac{(3x-1)^5}{15}+C$

b. $\frac{(3x-1)^5}{5}+C$

c. $\frac{(3x-1)^3}{3}+C$

d. $12(3x-1)^3+C$



2 Evaluate $\int (e^{2x}+\sin 3x)\,dx$.

a. $\frac{1}{2}e^{2x}-\frac{1}{3}\cos 3x+C$

b. $\frac{1}{2}e^{2x}+\frac{1}{3}\cos 3x+C$

c. $2e^{2x}-3\cos 3x+C$

d. $e^{2x}-\cos 3x+C$

3 A curve passes through $(1,4)$ and satisfies $\frac{dy}{dx}=4x-1$. Find the equation of the curve.

a. $y=2x^2-x+3$

b. $y=2x^2-x+4$

c. $y=4x^2-x+1$

d. $y=2x^2+x+1$

Section 2: Definite Integrals and Area Under Curves

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4 Evaluate the exact value of the definite integral $\int_1^4 \left(2x + \frac{1}{\sqrt{x}}\right) dx$.

a. 17

b. 15

c. 19

d. 16

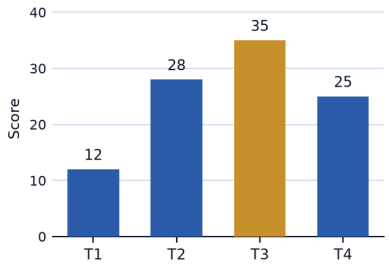
5 Find the area enclosed by the curve $y = 6x - x^2$ and the x-axis.

a. 36

b. 18

c. 72

d. 24



6 Evaluate $\int_0^{\frac{\pi}{4}} \sec^2 x \, dx$.

a. 1

b. $\sqrt{2}$

c. $\frac{1}{2}$

d. $\pi/4$

Section 3: Area Between Two Intersecting Curves

7 Find the area of the finite region enclosed between the parabola $y = x^2$ and the line $y = 2x$.

a. $\frac{4}{3}$

b. $\frac{2}{3}$

c. 2

d. $\frac{8}{3}$

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8 Find the area between the curves $y = 4 - x^2$ and $y = x^2 - 4$.

a. $\frac{64}{3}$

b. $\frac{32}{3}$

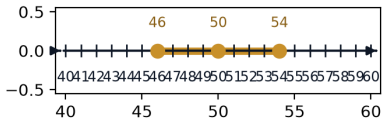
c. 16

d. 8

9 Set up and evaluate the integral representing the area between $y = \sqrt{x}$ and $y = x$ for $0 \le x \le 1$.

a. $\frac{1}{6}$

b. $\frac{1}{3}$



c. $\frac{1}{2}$

d. $\frac{2}{3}$

Section 4: Rectilinear Motion and Kinematics

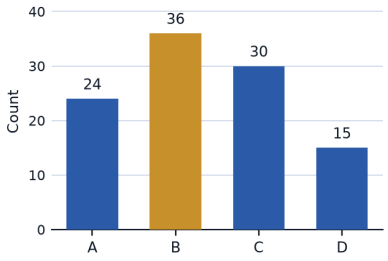
10 A particle moves in a straight line with velocity $v(t) = 3t^2 - 12t + 9$ m/s. Find the displacement from $t = 0$ to $t = 3$.

a. 0 m

b. 9 m

c. 18 m

d. 6 m



11 Find the acceleration of the particle in question 10 at $t = 4$ seconds.

a. 12 m/s^2

b. 6 m/s^2

c. 24 m/s^2

d. 0 m/s^2

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12 Calculate the total distance travelled by the particle with $v(t) = 3(t-1)(t-3)$ from $t = 0$ to $t = 3$.

a. 8 m

b. 0 m

c. 4 m

d. 12 m

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